UNIT IV – Mechanical & Electrical Energy Based Processes

23A03501 MACHINING PROCESSES · B.Tech Mechanical · JNTUK

I. Non-Traditional Machining (NTM) – Intro

NTM = material removed by **mechanical / thermal / chemical / electrical energy** directly – **no sharp cutting tool**, no direct tool-work hard contact.

Need for NTM

- Very hard materials (>400 HV): carbides, ceramics, HSS, Ti alloys
- Complex shapes: deep holes, micro-holes, 3D cavities, sharp corners
- Brittle / fragile jobs (glass, ceramics) – no cutting force
- Better surface finish & no burr / residual stress needed
- Tool material need NOT be harder than work

Classification (by energy source)

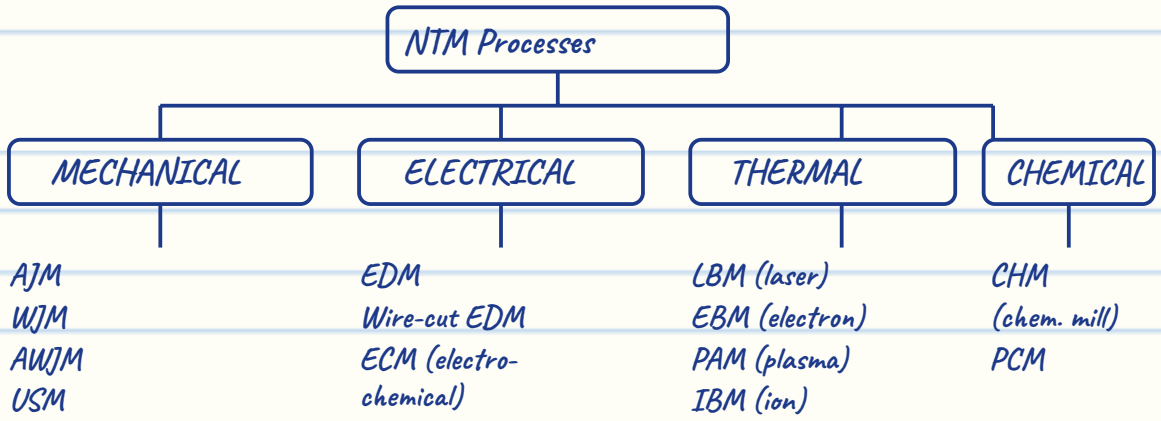


Fig 1. Classification of NTM by energy

NTM vs Conventional

Point	Conventional	NTM
Tool	Hard, sharp tool	Abrasive / jet / spark / wire
Work hardness	Limited by tool	Any hard material
Cutting force	High	Little / none
MRR	High	Low
Cost / power	Low	High
Accuracy, finish	Moderate	Very good

2. Abrasive Jet Machining (AJM)

Principle: high-velocity **gas jet ($air/N_2/CO_2$) + fine abrasive** particles strike work → **micro-chipping / erosion** (brittle fracture). Mechanical energy process.

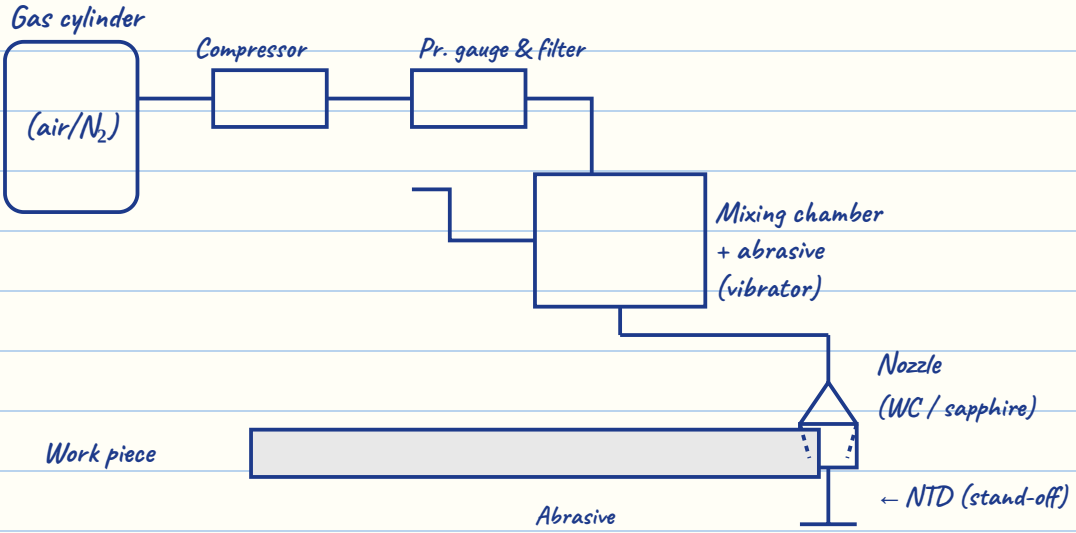


Fig 2. AJM equipment (gas → mixing chamber → nozzle → work)

Equipment

- Gas supply + compressor, pressure regulator, filter / dryer
- Mixing chamber with **vibrator** (keeps flow uniform)
- Nozzle: **WC or sapphire**, dia 0.2–0.8 mm (life 12–300 h)
- Work table / fixture, dust collector (enclosure)

Process parameters

Parameter	Typical value
Abrasive	Al_2O_3, SiC (also dolomite, glass beads)
Grain size	10–50 µm (fine); never re-use
Abrasive flow rate	3–20 g/min
Gas pressure	2–8 bar (kg/cm^2)
Jet velocity	150–300 m/s
Nozzle dia	0.2–0.8 mm
NTD / stand-off	0.25–15 mm (usually 1–2 mm)
Jet angle	60°–90°

Effect of parameters

- ↑ Pressure, ↑ flow rate → ↑ MRR (up to a limit; too much abrasive → collision loss)
- ↑ Stand-off → jet spreads → **taper, overcut** ↑, MRR first ↑ then ↓
- Finer grain → better finish, lower MRR

- Brittle work: MRR ∝ (vol per particle) ∝ $d^3.(v^2)^{(3/4)}$; ductile → low MRR

MRR model (Ghosh & Mallik – brittle, indentation)

- Each grain (dia d, velocity v, density ρ) indents hemispherical crater, width 2a
- Volume removed per impact ∝ $a^3 \propto d^3 (v\rho / H)^{(3/4)}$

$$Q = K \cdot Z \cdot d^3 \cdot (\rho v^2 / H)^{(3/4)}$$

- Z = no. of impacts/s, H = flow stress (hardness) of work, K = const
- Q ∝ abrasive flow rate (Z) → also Q ∝ v^1.5 → pressure matters

Example 1. Glass AJM: $Z = 2\times10^4$ impacts/s, volume removed per impact = 5×10^{-7} mm^3 → MRR = Z × vol = $2\times10^4 \times 5\times10^{-7}$ = **0.01 mm^3/s = 0.6 mm^3/min**. Doubling v (×2) → MRR × 2^1.5 = **2.83 times**.

Applications / Adv / Limits

- **Uses:** cutting, drilling, deburring, cleaning, etching of glass, ceramics, Ge, Si, mica; frosting; fine slots
- **Adv:** no heat damage, no vibration, can cut thin / brittle parts, low capital cost, complex shapes
- **Lim:** low MRR, **taper** in holes, abrasive embedding, nozzle wear, dust hazard, not for soft / ductile metals

3. Water Jet Machining (WJM)

Principle: **very high pressure water jet** (up to **4000 bar**) through tiny orifice → jet velocity ≈ 900 m/s (≈ 2.5 × sound speed) → **kinetic energy → high stress at work → erosion / cutting** of soft materials. (Pure water, no abrasive.)

Jet velocity (Bernoulli)

$$\tfrac{1}{2}\rho v^2 = p \Rightarrow v = \sqrt{(2p/\rho)} \; ; \; (\times Cd \text{ for losses})$$

- Eg: p = 4000 bar = 4×10^8 Pa, ρ = 1000 → $v = \sqrt{(8\times10^5)} \approx$ **894 m/s**
- Q (flow) = Cd · A · v ; Jet power = $\tfrac{1}{2}\dot{m} v^2 = p \cdot Q$

Equipment / parameters

- **Intensifier** (hydraulic oil piston, area ratio ~20:1) → 1500–4000 bar
- **Orifice (nozzle):** sapphire / ruby / **diamond**, dia 0.1–0.35 mm; life 100–200 h (diamond ~1000 h)
- Stand-off 2–6 mm; traverse speed 0.5–10 m/min
- Additives: long-chain polymers → coherent jet
- Catcher: absorbs jet energy; filters, accumulator for pulsation damping

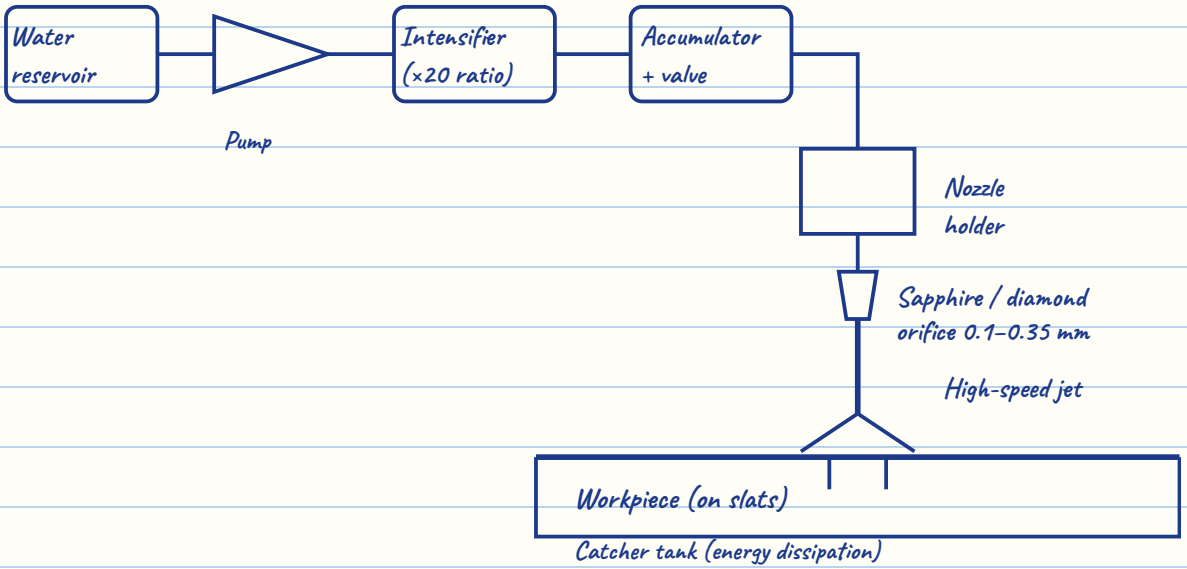


Fig 3. Water jet machine – intensifier circuit

- **Materials:** rubber, foam, paper, plastics, leather, food, wood, carpet, gasket, composites (soft)
- **Adv:** no heat (cold cut), no tool wear, narrow kerf, no dust, any contour, clean / hygienic, minimal burr
- **Lim:** cannot cut hard metals/ceramics, high capital, noise (≈ 90 dB), taper on thick jobs, kerf wanders

4. Abrasive Water Jet Machining (AWJM)

Principle: WJM + **abrasive** particles (garnet) entrained into high-pressure jet → **abrasive erosion** → cuts metals, ceramics, composites, stone. Abrasive jet speed ≈ 300–600 m/s.

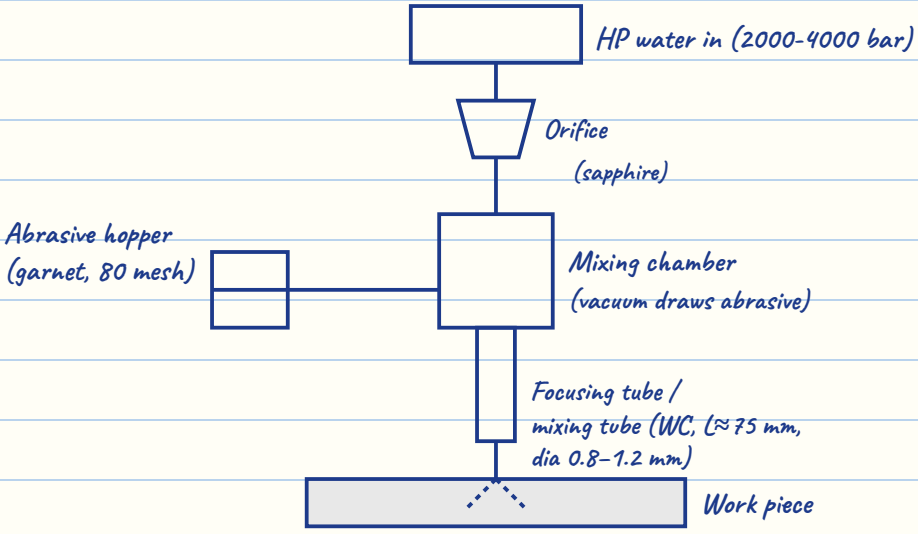


Fig 4. AWJM cutting head

- Abrasive: **garnet** (most common), Al_2O_3, olivine; mesh 60–120; flow 0.1–0.5 kg/min

- **Mixing tube** (focusing nozzle): WC / composite carbide, ID 0.76–1.2 mm, L 50–100 mm
- Orifice dia 0.25–0.4 mm; pressure 2000–4000 bar; stand-off 1–3 mm
- Mixing: **entrainment (suction type)** – common; or slurry (direct pumping) type
- Cutting speed ↑ → kerf taper, striations; ↓ speed → deeper cut
- Depth of cut $h \propto P^a \cdot \dot{m}_a^b / (V_{traverse})$ (↑ P, ↑ abrasive, ↓ speed)
- **Uses:** steel, Ti, Inconel, Al, granite, marble, glass, ceramics, CFRP; aerospace panels, tile / stone cutting
- **Adv:** no HAZ, cuts almost any material up to 200 mm thick, no thermal distortion, CNC any shape
- **Lim:** kerf taper, rough (striations), costly abrasive & disposal, nozzle (mixing tube) wear, noise, slow vs laser

WJM vs AWJM

Point	WJM	AWJM
Medium	Pure water	Water + garnet
Work	Soft, non-metals	Metals, ceramics, hard
Nozzle	Orifice only	Orifice + mixing tube
Jet dia	0.1–0.35 mm	0.8–1.2 mm

5. Ultrasonic Machining (USM)

Principle: tool vibrates at **ultrasonic frequency (≈ 20–25 kHz)**, small amplitude (15–50 µm), perpendicular to work; **abrasive slurry** between tool and work. Abrasive grains hammered → micro-chipping of brittle work. Tool shape → **mirror image** cavity.

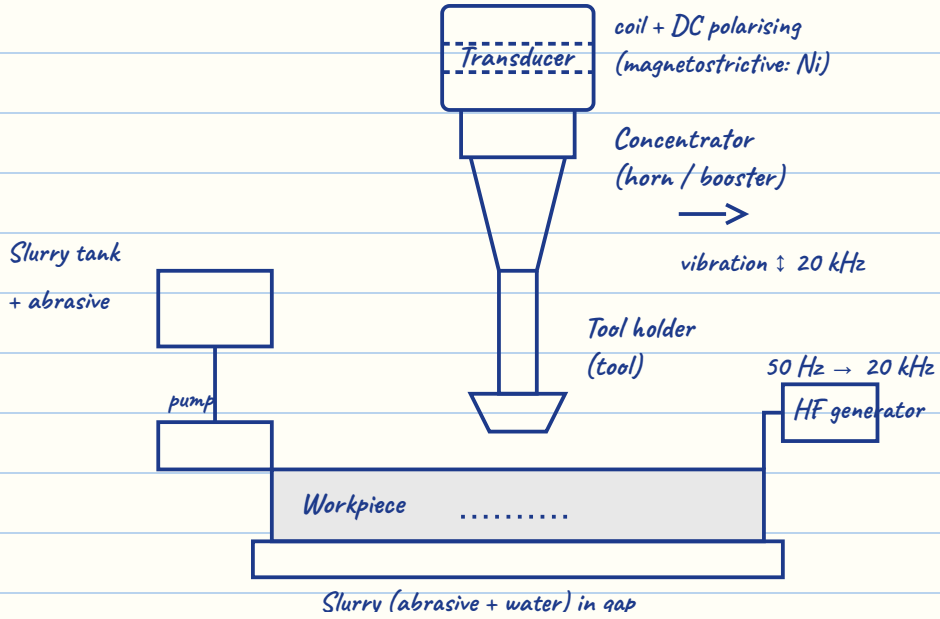


Fig 5. Ultrasonic machine

Equipment

- **HF generator:** 50 Hz mains → 20 kHz electrical signal
- **Transducer:** electrical → mechanical vibration
  - **Magnetostrictive** (Ni, Permalloy): length changes in magnetic field; ΔL = 20–30 µm; robust, efficiency ~25 %, cooling needed
  - **Piezoelectric** (quartz, PZT): efficiency 90 %, less amp., used for low power
- **Concentrator (horn):** amplifies amplitude (exponential / stepped / conical); material Monel, Ti, steel; length = $\lambda/2$ = half-wave resonance
- **Tool:** soft-tough (brass, mild steel, SS), tapered; wear ~ 1:100 (hard work)
- **Slurry:** abrasive : water ≈ 1:1 (30–40 % grit); circulated, cools & flushes debris

Horn length $L = \lambda/2 = c / 2f$ ($c = \sqrt{(E/\rho)}$)

Process parameters

Parameter	Value / effect
Frequency f	19–25 kHz (resonance)
Amplitude a	15–50 µm; MRR ↑ with a
Abrasive	B_4C (best, costly), SiC, Al_2O_3, diamond
Grit size d	#100–#800 (10–100 µm); coarse → ↑ MRR
Slurry conc. C	20–40 % by volume
Static feed force F	Optimum exists (max MRR)
Work hardness	↑ H → ↓ MRR

MRR – Shaw's model (assumptions)

- Abrasive grains spherical dia d; work brittle → crater is hemisphere, dia D (chip)

- Volume removed/grain per cycle $\propto D^3$; from geometry $D \approx 2\sqrt{(d \cdot h)}$, h = indentation depth
- So vol/grain $\propto (d \cdot h)^{(3/2)}$; N grains hit per cycle ($N \propto C/d^2$)
- f cycles/s → **$MRR = K \cdot N \cdot (d \cdot h)^{(3/2)} \cdot f$**

$$Q = K \cdot N \cdot f \cdot (a \cdot d)^{(3/2)} \Rightarrow Q \propto f \cdot a^{(3/2)} \cdot d^{(3/2)}$$

- Conclusion: MRR $\propto$ **f, $a^{(3/2)}$, $d^{(3/2)}$**, and ↓ with work hardness; Q = vol/unit time
- Typical MRR: glass ≈ 5–15 mm^3/min (depends on tool area); K absorbs work hardness H

Example 2. USM, f = 20 kHz, vol per grain-hit = 1×10^{-6} mm^3, grains hitting per cycle = 10 →
$Q = 20000 \times 10 \times 10^{-6}$ = **0.2 mm^3/s = 12 mm^3/min**. If amplitude is raised ×1.5 → $Q \times 1.5^{1.5} = 1.84$ → **0.37 $mm^3/s \approx$ 22 mm^3/min.**

Applications / Adv / Limits

- **Uses:** glass, quartz, ceramics, diamond, WC dies, Ge/Si wafers; drilling, engraving, wire-drawing dies, cutting gems, non-round holes
- **Adv:** works non-conductive brittle materials, no heat / stress, no burr, any shape hole, good finish (0.2–0.8 µm Ra)
- **Lim:** **low MRR**, high tool wear, deep holes hard (slurry access), unsuitable for soft ductile metals, expensive setup, limited size of hole

6. Comparison: AJM · WJM · AWJM · USM

Item	AJM	WJM	AWJM	USM
Medium	Gas+abrasive	Water	Water+garnet	Slurry+tool
Mechanism	Erosion	Erosion/shear	Abrasive erosion	Hammering, chipping
Pressure/Freq	2–8 bar	≤ 4000 bar	≤ 4000 bar	20 kHz
Tool	Nozzle	Sapphire orifice	Orifice+mixing tube	Shaped tool
Work	Brittle thin	Soft	Almost all	Hard brittle
MRR	Very low	High (soft)	Medium	Low
Finish	Good	Fair	Rough	Very good
HAZ	None	None	None	None

7. Electric Discharge Machining (EDM)

EDM (spark erosion / spark machining): material removed by **repeated electric sparks** between tool (electrode) & work, both **immersed in dielectric**, gap 0.01–0.5 mm. Work must be **electrically conductive**. Thermal process. (Invented by Lazarenko, 1943).

Working principle (mechanism)

1. Voltage applied across gap (60–300 V); electrode (–) work (+) usually
2. Ionisation: field ≈ 10^5–10^6 V/mm → dielectric breaks down at shortest gap
3. **Plasma channel** forms, current flows, **spark** (temp 8000–12000 °C, 10^8 W/mm^2)
4. Local melting + vaporisation of tiny crater (≈ 1–100 µm)
5. Pulse OFF → plasma collapses, dielectric flushes molten globules (debris)
6. Servo moves tool → gap kept constant; cycle repeats 2000–500000 /s

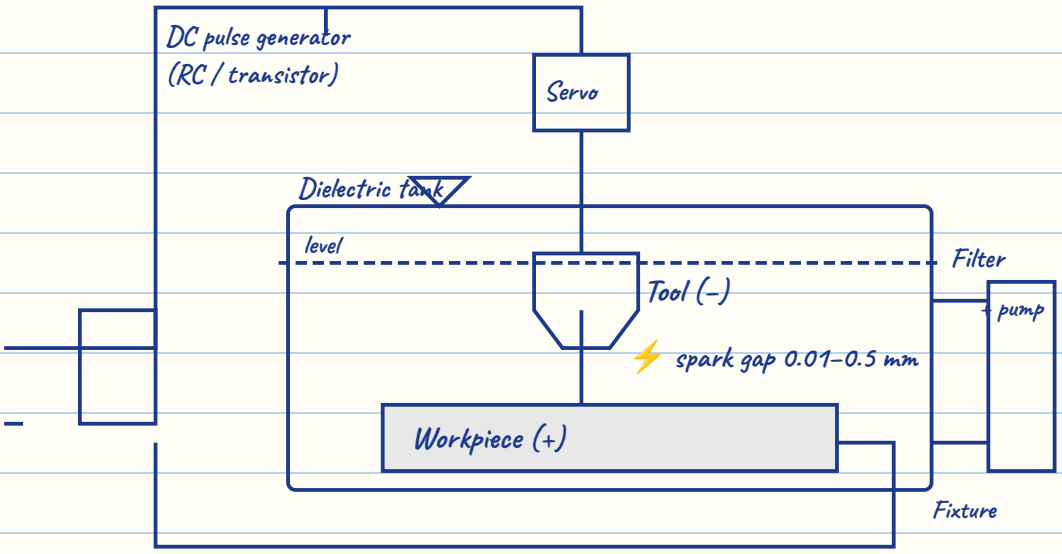


Fig 6. EDM schematic

Main elements

- **Power supply:** DC pulse (RC relaxation / transistor-pulse / RLC)
- **Servo feed system:** keeps gap constant, retracts on short-circuit
- **Dielectric system:** tank, pump, filter (≈ 5 µm), heat exchanger
- Electrode (tool) + work holder, table (X-Y)

Dielectric fluid & functions

- Types: **kerosene, EDM oil (hydrocarbon), transformer oil, deionised water** (wire-EDM), glycol
- Functions: (1) insulates till breakdown (2) **ionises** (3) **flushes** debris (4) **cools** tool/work (5) constricts spark → ↑ energy density

- Req: high dielectric strength, low viscosity, high flash point, non-toxic, cheap

Flushing methods

- **Pressure flushing** through tool hole (most common)
- **Suction flushing** (vacuum through work/tool) – less taper
- **Side / jet flushing** – nozzle directs fluid at gap
- **Pulsed / jump flushing** – tool lifts periodically; **rotating electrode**

8. RC Relaxation Circuit (Lazarenko)

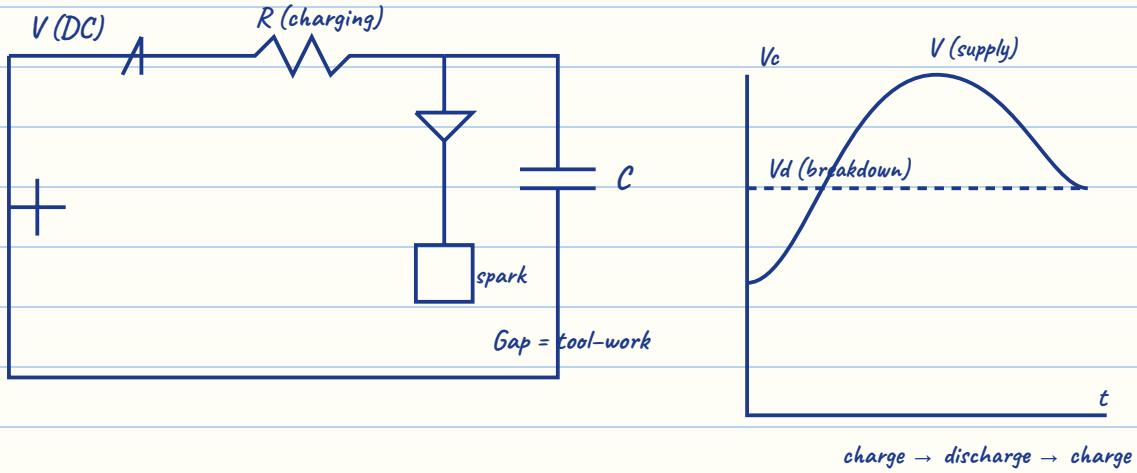


Fig 7. RC circuit & capacitor voltage cycle

- R charges C from V; when Vc = **Vd (breakdown)** → spark → C discharges → repeats
- R ≫ spark-gap resistance, so C discharges only through gap

Derivation

$$V_c = V\,(1 - e^{(-t/RC)})$$

- Time to reach Vd : $t_c = RC \cdot \ln[V / (V - V_d)]$
- Discharge time $t_d \ll t_c$, so **cycle time $T \approx t_c$**; frequency $f = 1/t_c$
- Energy per spark: $E = \frac{1}{2} C V_d^2$ → Power $P = E \cdot f = \frac{1}{2} C V_d^2 f$

$$P = \frac{1}{2} C V_d^2 / [RC \ln(V/(V-V_d))] = \frac{1}{2} V_d^2 / [R \ln(V/(V-V_d))]$$

- Let $V_d = x \cdot V$ → $P = (V^2/2R) \cdot x^2 / \ln[1/(1-x)]$
- For max P: $dP/dx = 0$ → solve gives **$x = 0.716 \approx 0.72$**

★ Important

- Max power (max MRR) when **Vd = 0.72 V** (break-down at 72 % of supply)
- MRR ∝ energy/spark × freq ; **MRR ≈ 40·I·Tm^(-1.23)** (empirical, mm^3/min, I amp, Tm melt pt °C)
- Finish ↑ with small C (low energy) but MRR ↓

Example 3. V = 200 V, R = 50 Ω, C = 10 µF, Vd = 144 V (=0.72V).

$t_c = RC \ln[V/(V-Vd)] = 50\times10\times10^{-6} \times \ln(200/56) = 5\times10^{-4} \times 1.273 =$ **0.636 ms**

$f = 1/t_c =$ **1571 Hz** ; $E = ½CVd^2 = ½\times10^{-5}\times144^2 =$ **0.1037 J**

$P = E\cdot f = 0.1037 \times 1571 \approx$ **163 W**

Example 4. Same circuit; if specific removal ≈ 0.02 mm^3 per joule (assumed for steel) → **MRR ≈ 0.02 × 163 ≈ 3.3 mm^3/s ≈ 196 mm^3/min.**

Other pulse generators

- **Transistor (controlled pulse):** square pulse, set on-time (Ton), off-time (Toff), current; modern, ↓ tool wear, high MRR
- RC: simple, slow, rough finish; used for micro-EDM, finishing

9. EDM Process Parameters

Parameter	Effect
Pulse-on time Ton (1–2000 µs)	↑ Ton → ↑ MRR, ↑ crater, ↑ Ra, ↑ HAZ, ↑ overcut
Pulse-off time Toff	↑ → better flushing, stable; ↓ MRR
Peak current Ip (1–100+ A)	↑ MRR (∝ Ip), ↑ roughness, ↑ tool wear
Gap voltage (40–100 V)	↑ gap, ↑ overcut
Open-circuit V (60–300 V)	Breakdown / gap width
Polarity	Tool –ve (work +) → max MRR; tool +ve for graphite & low wear
Flushing pressure	Debris removal, stability

Duty cycle $\eta = Ton / (Ton + Toff) \times 100\ \%$

- Pulse frequency f = 1/(Ton+Toff); higher η → higher MRR but arcing risk
- Typical: MRR 5–500 mm^3/min; Ra 0.2–12 µm; accuracy ±0.05 mm; gap 0.01–0.5 mm

Electrode (tool) materials

- **Copper** – good conductivity, fine finish, tool wear low

- **Graphite** – cheap, easy to machine, high MRR, sublimes, most used
- **Brass** – easy to make, high wear; micro-holes, **Cu-W / Ag-W** – lowest wear, costly, for carbides
- Req: high melting point, conductivity, easily machined, low wear

Tool wear, overcut, surface

- **Tool wear ratio (TWR)** = vol. tool removed / vol. work removed (graphite ~1–10 %, Cu 2–10 %, Cu-W ~ 1 %)
- Wear: corner wear > end wear > volume wear; wear ∝ 1/melting pt
- **Overcut** = (hole size – tool size)/2 = spark gap; **about 0.01–0.5 mm**; gives tool size = cavity – 2 × gap
- **Surface:** crater overlap → matt finish; Ra ↑ with Ip, Ton

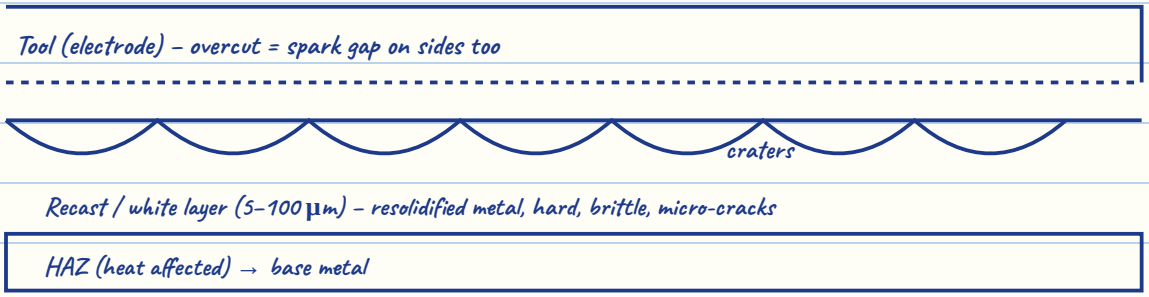


Fig 8. Machined surface: craters, recast layer, HAZ

- **Recast layer:** molten metal re-solidified; hard, micro-cracks, tensile residual stress → ↓ fatigue; removed by polishing / finishing pass
- HAZ below it: heat-affected zone with altered structure

Applications / Adv / Limits

- **Uses:** forging / extrusion / press dies, moulds, punches, turbine blades, cooling holes, deep micro-holes, tool-steel & carbide cutting, broken tap removal
- **Adv:** any conductive hard material, complex 3D cavities, no cutting force, thin delicate parts, good accuracy, burr-free, can be automated (CNC)
- **Lim:** only **conductive** work, low MRR, **tool wear**, recast layer / HAZ, sharp internal corners difficult, high power / cost, fire hazard (oil), taper

10. Wire-Cut EDM (WEDM)

Principle: same spark erosion but tool = **continuously moving thin wire** (like a band saw) – **travelling wire** cuts the work along a programmed (CNC) path in **deionised water**. No contact; wire is used once.

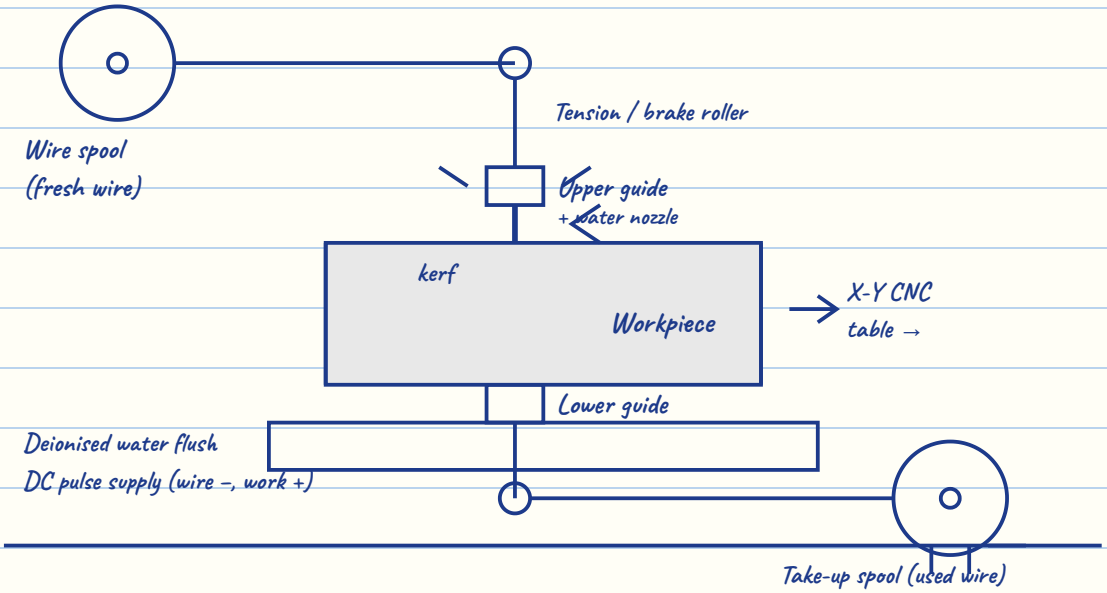


Fig 9. Wire-cut EDM

Features / parameters

- **Wire:** brass (Cu 63 / Zn 37), **zinc-coated brass**, copper, Mo, tungsten (fine); dia **0.05–0.3 mm** (0.25 usual); tensile strength ~ 900 N/mm^2; speed 1–15 m/min, used once
- **Wire guides:** sapphire / diamond / ceramic; upper & lower; give accuracy ±0.005 mm
- **Spool, tension & wire feed system:** constant tension 5–20 N; wire cassette
- **CNC X-Y table** (+ U-V upper guide for **taper cutting** up to ±30°)
- **Dielectric: deionised water** (resistivity 10^4–10^5 Ω·cm) – jets coaxial with wire, with ion-exchange resin filter
- Gap ≈ 0.025–0.05 mm; kerf = wire dia + 2×gap; Ton 0.1–25 µs; Ip 100–500 A (peak)
- Wire path programmed: **G-code**; start hole needed (or EDM drilled); "skim cuts" for finish
- Cutting speed (mm^2/min) = feed × job thickness; 100–300 mm^2/min; Ra 0.3–2.5 µm

$$MRR = \text{kerf width} \times \text{job height} \times \text{cutting speed}$$

Example 5. Wire 0.25 mm, gap 0.03 mm each side, height 40 mm, feed 3 mm/min → kerf = 0.25 + 2×0.03 = 0.31 mm → **MRR = 0.31 × 40 × 3 = 37.2 mm^3/min.**

- **Uses:** punches, dies, stripper plates, extrusion dies, gears, templates, turbine slots, medical parts, PCD / carbide tools, prototype

- **Adv:** no special tool shape (no electrode making), complex profiles, high precision, no wear (wire fresh), unattended running, tapered/ 2½D, hard materials
- **Lim:** only through (profile) cuts, conductive material, slow, high initial & running cost (wire), recast layer, thick sections slow

EDM (die-sinking) vs Wire EDM

Point	Die-sinking EDM	Wire EDM
Tool	Shaped electrode (Cu/graphite)	Moving wire (0.05–0.3 mm)
Dielectric	Kerosene / EDM oil	Deionised water
Tool wear	Significant	Negligible (fresh wire)
Shapes	Blind cavities, 3D	Through, profile cuts
Tool cost	High (making electrode)	No tool making
Flushing	Pressure / suction	Water jets coaxial
Typical use	Moulds, forging dies	Punches, dies, gears

II. Quick Revision & Likely Questions

★ *Important*

- AJM: gas 2–8 bar, 10–50 µm Al_2O_3/SiC, 150–300 m/s, nozzle WC/sapphire, NTD 1–2 mm
- WJM: ≤ 4000 bar, $v \approx \sqrt{(2p/\rho)} \approx$ 900 m/s, sapphire/diamond orifice 0.1–0.35 mm
- AWJM: + garnet; mixing (focusing) tube WC; cuts hard metals, no HAZ
- USM: 20 kHz, 15–50 µm amp.; B_4C/SiC/Al_2O_3 slurry; MRR ∝ f·a^1.5·d^1.5; transducer + horn
- EDM: spark erosion in dielectric; $E = \frac{1}{2}CV^2$; **Vd = 0.72 V** for max power; MRR↑ with Ip, Ton
- WEDM: brass wire 0.05–0.3 mm, DI water, CNC, taper cutting, no tool wear

Mnemonics

- NTM energy: **M-E-T-C** = Mechanical, Electrical, Thermal, Chemical
- EDM fluid functions: **I-I-F-C** – Insulate, Ionise, Flush, Cool
- USM chain: Generator → Transducer → Horn → Tool → Slurry → Work

Likely questions

1. Classify NTM; compare with conventional machining. (5 M)
2. With a neat sketch explain AJM; process parameters, merits & demerits. (10 M)
3. Explain WJM and AWJM; differences and applications. (10 M)
4. Describe USM with neat sketch; derive MRR (Shaw). Transducers? (10 M)
5. Principle of EDM; dielectric & flushing; electrode materials. (10 M)
6. Derive for RC circuit: f, E, and show optimum Vd = 0.72 V. (10 M)
7. Wire-cut EDM: working, parameters, applications; compare with EDM. (10 M)
8. Short notes: recast layer, overcut, tool wear, duty cycle, horn, garnet, taper cutting. (2–5 M)
9. Numericals: RC-circuit spark energy, frequency, power; MRR of USM/AJM.